Scratch and Peck Feeds

AI for Marketing Week 4

Using AI to Generate Video
RAG: Retrieval-Augmented Generation
Competitor Analysis Project
Projects

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Week 3 2026

Agenda

- **Using AI to Create Video**
  - AI Video Generators
  - Creating Prompts for Video
    - Features
    - Timing
- **Retrieval Augmented Generation**
  - What is RAG and why use it?
  - How does RAG Work
  - RAG Demo
- **Corporate AI Use Policy**
  - Foundation and Access
  - Security, Usage, Accessibility
- **Projects**
  - Ideas for a Project
  - Getting Help with the Project

Using AI to Create Video

Popular AI Video Generators

- Google (Veo 3.1)
- Kling
- Runway ML
- Invidio
- Luma
- Firefly
- Invideo

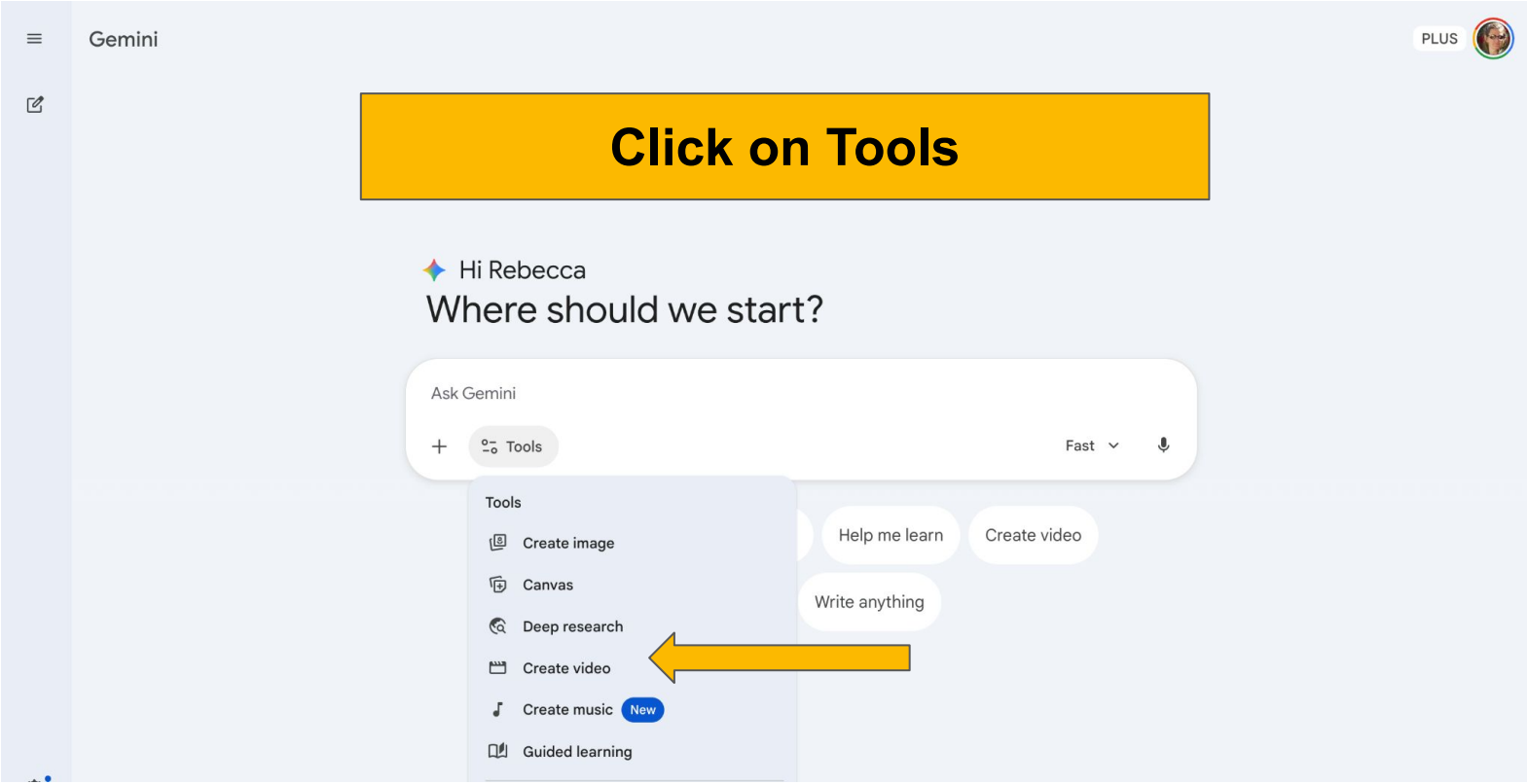
Creating Prompts for Video

Subject	Specify main focus of the video: a person communicating, an animal, a garden
Action	Describe what is happening: animal is making animal sounds, person is climbing a mountain, cartoon figure disappears into smoke
Setting	Where does the video take place: under water, out in space, in a barn
Style	Choose from a variety of styles like: cinematic, 3D animation, lo-fi aesthetic, black and white film noir,hyper-realistic)
Camera	How camera moves throughout the video: drone shot, extreme close-up, slow zoom-in or zoom-out, panning from left to right for right to left
Mood	Use descriptive adjectives like sad, happy, high-energy, dreamy, tense

Length of Video

- most videos 2-10 seconds
- the model may ignore specifying a length of 10 minutes
- Not common to specify a length
- Focus of instruction should be on
- pacing:
  - longer/slower: Use **slow-motion**, **leisurely**, **lingering shot**, **sluggish**, **gradual**
  - faster/shorter: Use **high-speed**, **fast-paced**, **rapid**, **quick cuts**, **dynamic**
  - sequence of events

Using Google to Create Video



Sample Prompt

Generate a prompt that shows a chicken letting the world know that she has laid an egg. The camera can start from a short distance and zoom in on her. She will be making a characteristic sound so her beak will be moving. After a few seconds, she can get up and walk away to show her egg. The style should be cinematic. The mood is one of winning and demonstrating a champion.

Sample Video:

Cinematic Chicken Champion Egg Announcement

RAG: Retrieval Augmented Generation

Benefits to Using RAG

Managing Your Own LLM Locally Can Potentially

- Reduce cost for AI Access
- Reduce Security Issues
- Reduce Hallucinations
- Provide Prompt Context from Business Documents

Recommended Hardware

NVDIA GPU

or

Apple M1,M2,M3 & 32+ GB Memory

16 GB Memory

Recommended Software

Ollama

LMStudio

OpenWebUI

What is RAG Used For?

- **Add your own business content to the LLM**
  - **Content is converted to a database of numerical chunks that LLMs understand called Embeddings**
  - **User can query the LLM and get back content relevant to their business**
  - **RAG is hosted by the business**
- **Business controls who has access to your RAG LLM and Embeddings**

Sample Raw Documents (PDF format)

Standard Operating Procedure

Employee Onboarding

Document Number:	HR-SOP-001
Effective Date:	January 1, 2026
Last Revised:	January 1, 2026
Approved By:	Director of Human Resources

1. Purpose

This Standard Operating Procedure establishes the protocol for onboarding new employees to ensure a smooth transition into the organization and compliance with all company policies and procedures.

Standard Operating Procedure

Employee Recognition and Rewards Program

Document Number:	HR-SOP-004
Effective Date:	January 1, 2026
Last Revised:	January 1, 2026
Approved By:	Director of Human Resources

1. Purpose

This Standard Operating Procedure outlines the company’s comprehensive recognition and rewards program designed to acknowledge employee contributions, celebrate milestones, and foster a culture of excellence and community engagement.

Standard Operating Procedure

Employee Healthcare Benefits Program

Document Number:	HR-SOP-005
Effective Date:	January 1, 2026
Last Revised:	January 1, 2026
Approved By:	Director of Human Resources

1. Purpose

This Standard Operating Procedure outlines the comprehensive healthcare benefits available to eligible employees and their families, including enrollment procedures, coverage options, and program details.

Sick Leave Policy

Standard Operating Procedure

Effective Date: January 1, 2026
Policy Owner: Human Resources Department
Last Reviewed: January 1, 2026

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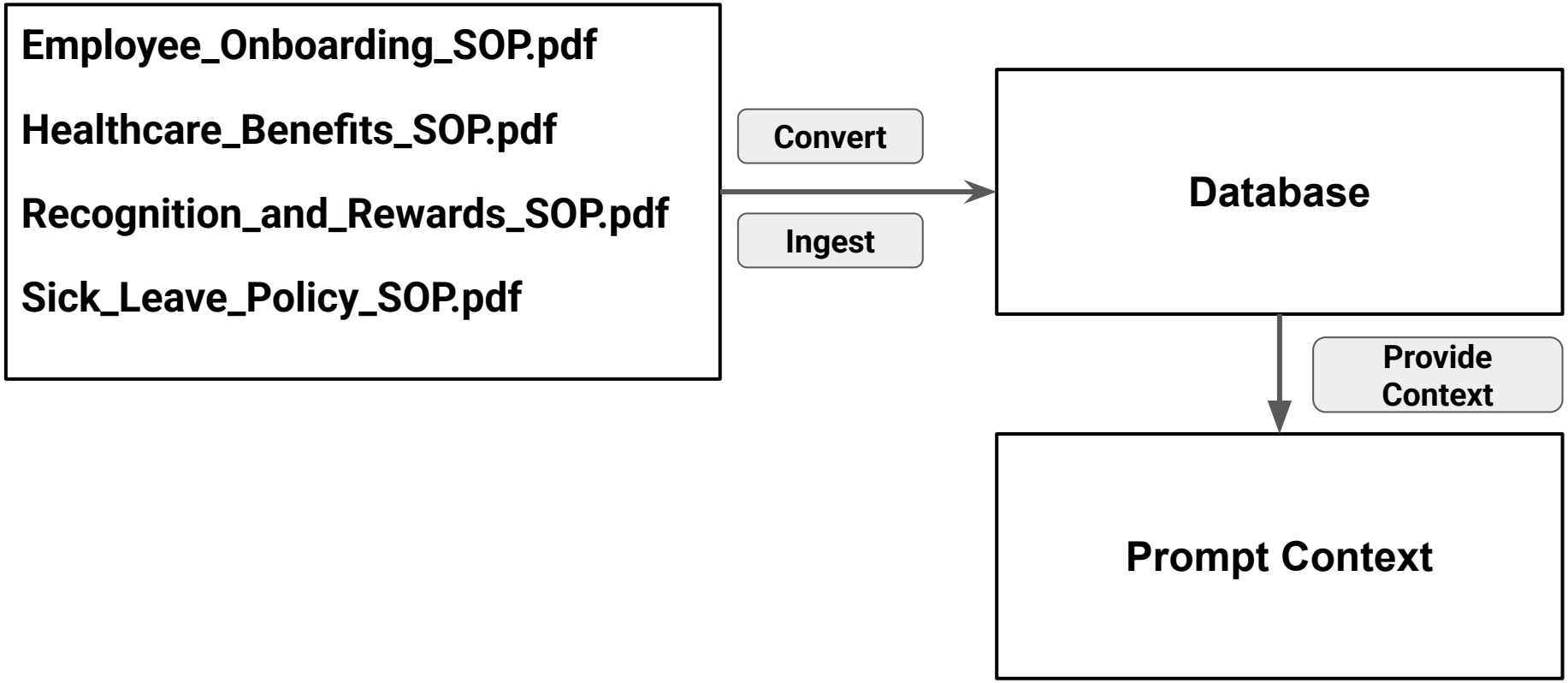
1. Purpose

This policy establishes guidelines for sick leave usage to ensure employees can address health needs while maintaining operational continuity.

Create an HR Database Using User Documents



Sample Processed Documents (Markdown format)

```
## Standard Operating Procedure Employee Onboarding

| Document Number:   | HR-SOP-001                  |
|--------------------|-----------------------------|
| Effective Date:    | January 1, 2026             |
| Last Revised:      | January 1, 2026             |
| Approved By:       | Director of Human Resources |

## 1. Purpose

This Standard Operating Procedure establishes the protocol for onboarding new employees to
ensure a smooth transition into the organization and compliance with all company policies and
procedures.
```

```
## Standard Operating Procedure

## Employee Healthcare Benefits Program

| Document Number:   | HR-SOP-005                  |
|--------------------|-----------------------------|
| Effective Date:    | January 1, 2026             |
| Last Revised:      | January 1, 2026             |
| Approved By:       | Director of Human Resources |

## 1. Purpose

This Standard Operating Procedure outlines the comprehensive healthcare benefits available to
eligible employees and their families, including enrollment procedures, coverage options, and
program details.
```

```
## Standard Operating Procedure

## Employee Recognition and Rewards Program

| Document Number:   | HR-SOP-004                  |
|--------------------|-----------------------------|
| Effective Date:    | January 1, 2026             |
| Last Revised:      | January 1, 2026             |
| Approved By:       | Director of Human Resources |

## 1. Purpose

This Standard Operating Procedure outlines the company&amp;amp;#x2019;s comprehensive recognition
and rewards program designed to acknowledge employee contributions, celebrate milestones, and
foster a culture of excellence and community engagement.
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```
## Sick Leave Policy

## Standard Operating Procedure

Effective Date:

January 1, 2026

Policy Owner:

Human Resources Department

Last Reviewed:

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## 1. Purpose

This policy establishes guidelines for sick leave usage to ensure employees can address health
needs while maintaining operational continuity.
```

RAG Prompt Template

```
template = """

You are a helpful assistant. Use the context below to answer the question.

If the context does not contain the answer, say: "I could not find that
information."

Context:

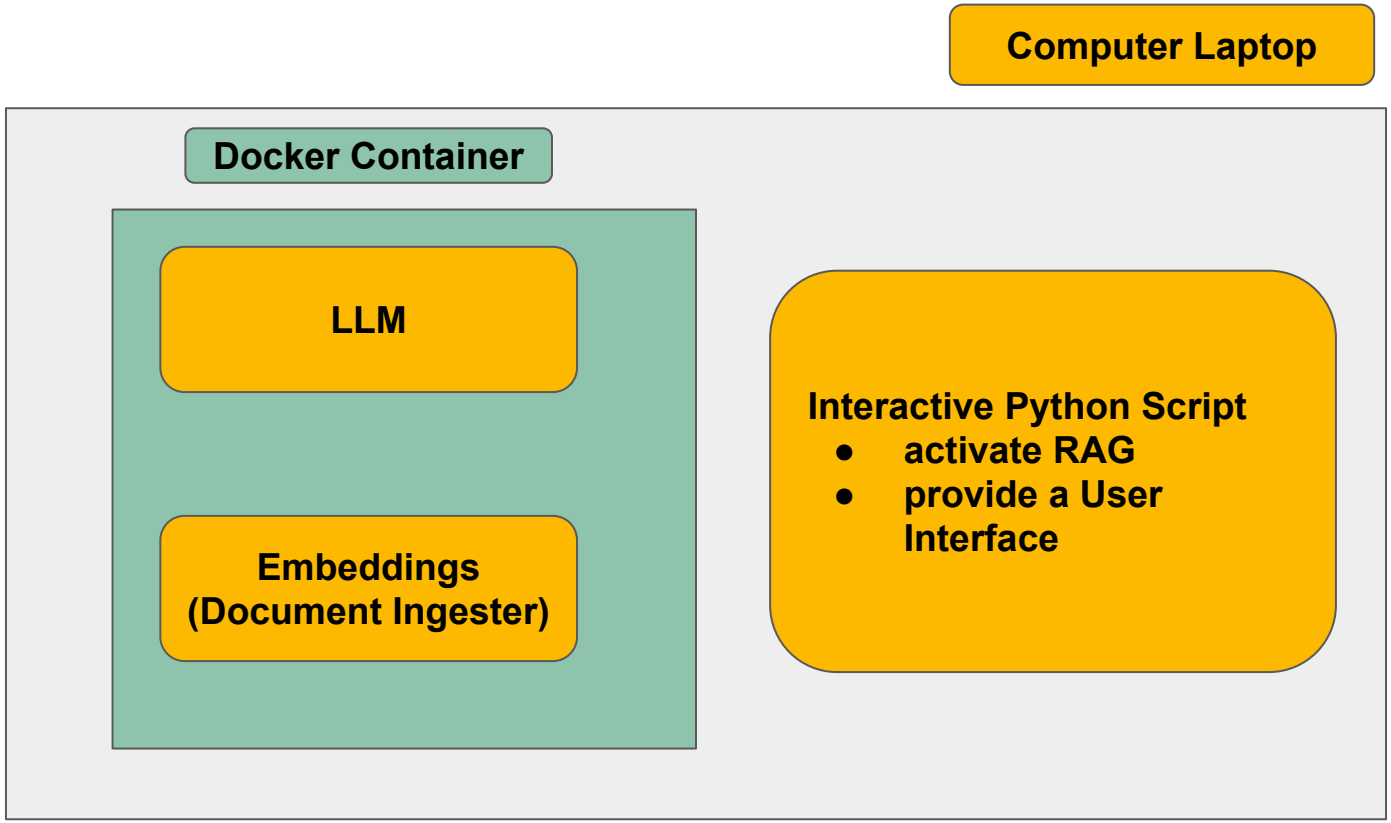
{context}

Question: {question}

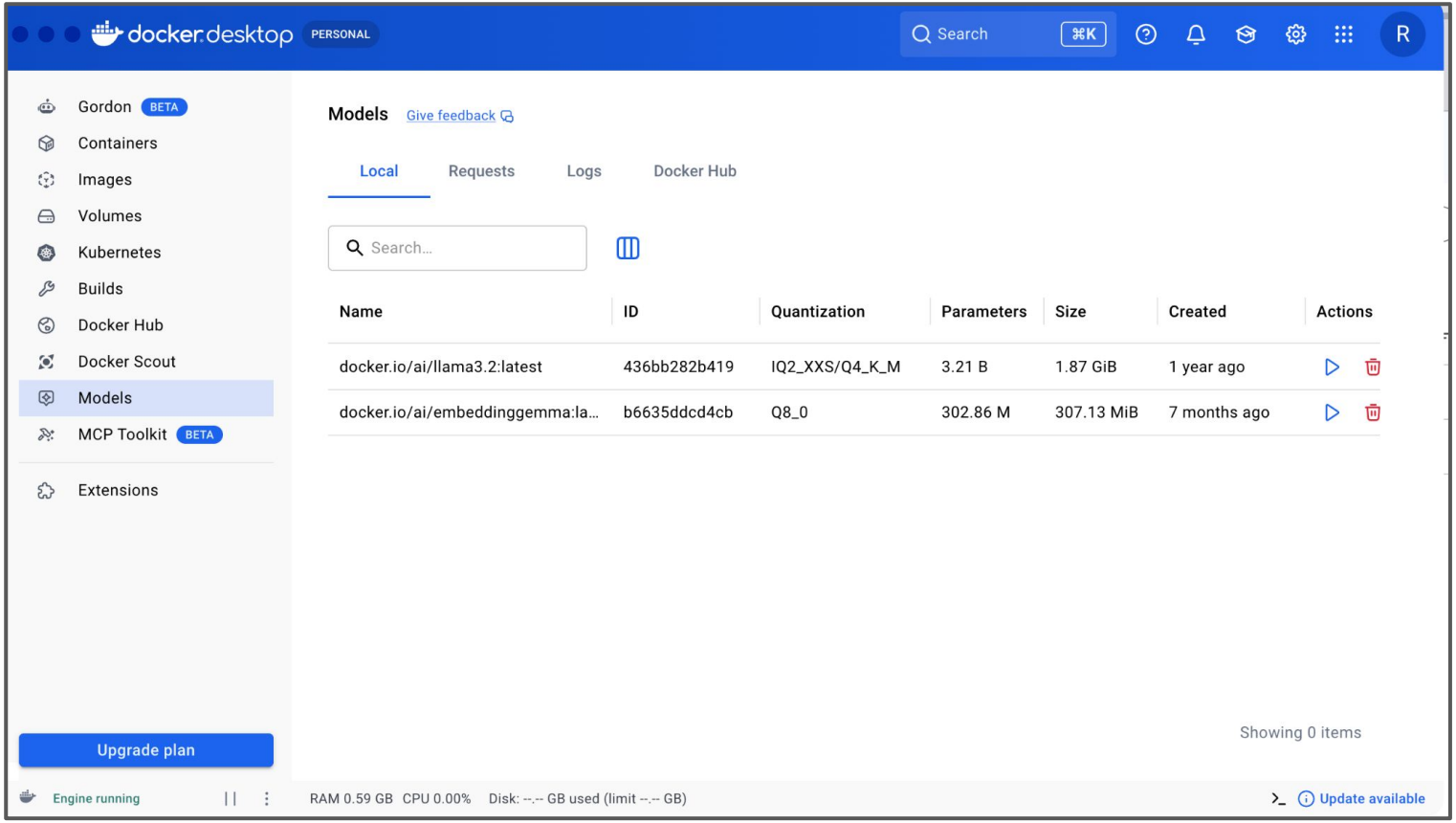
Answer:

"""
```

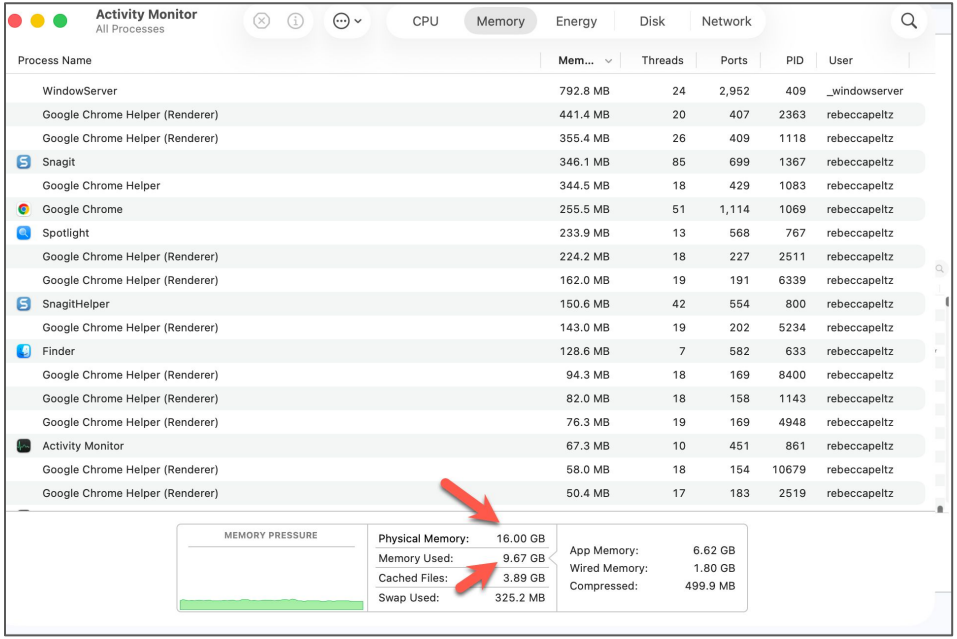
Simple Implementation of RAG on a Laptop



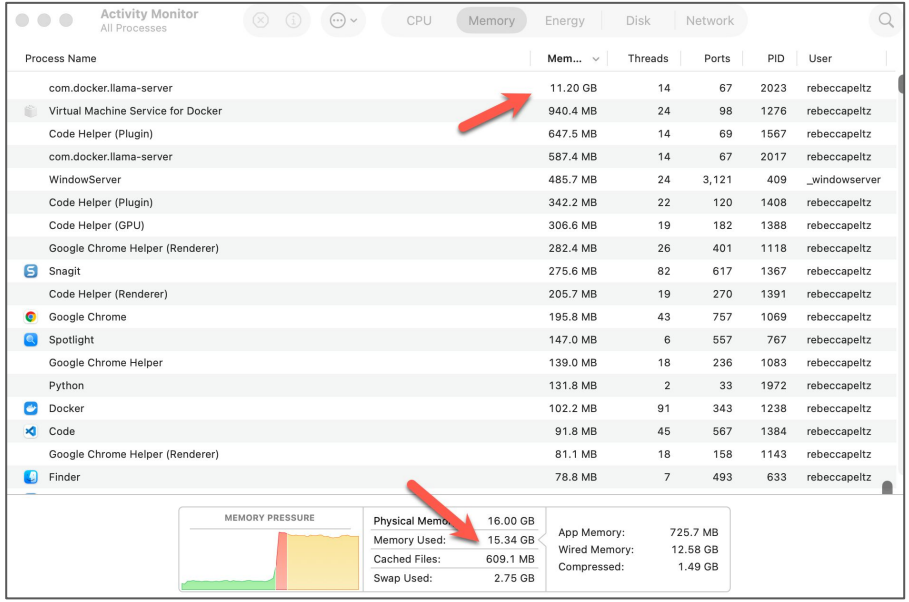
Docker Setup: 2 Models llama3 and embeddinggema



Memory Usage

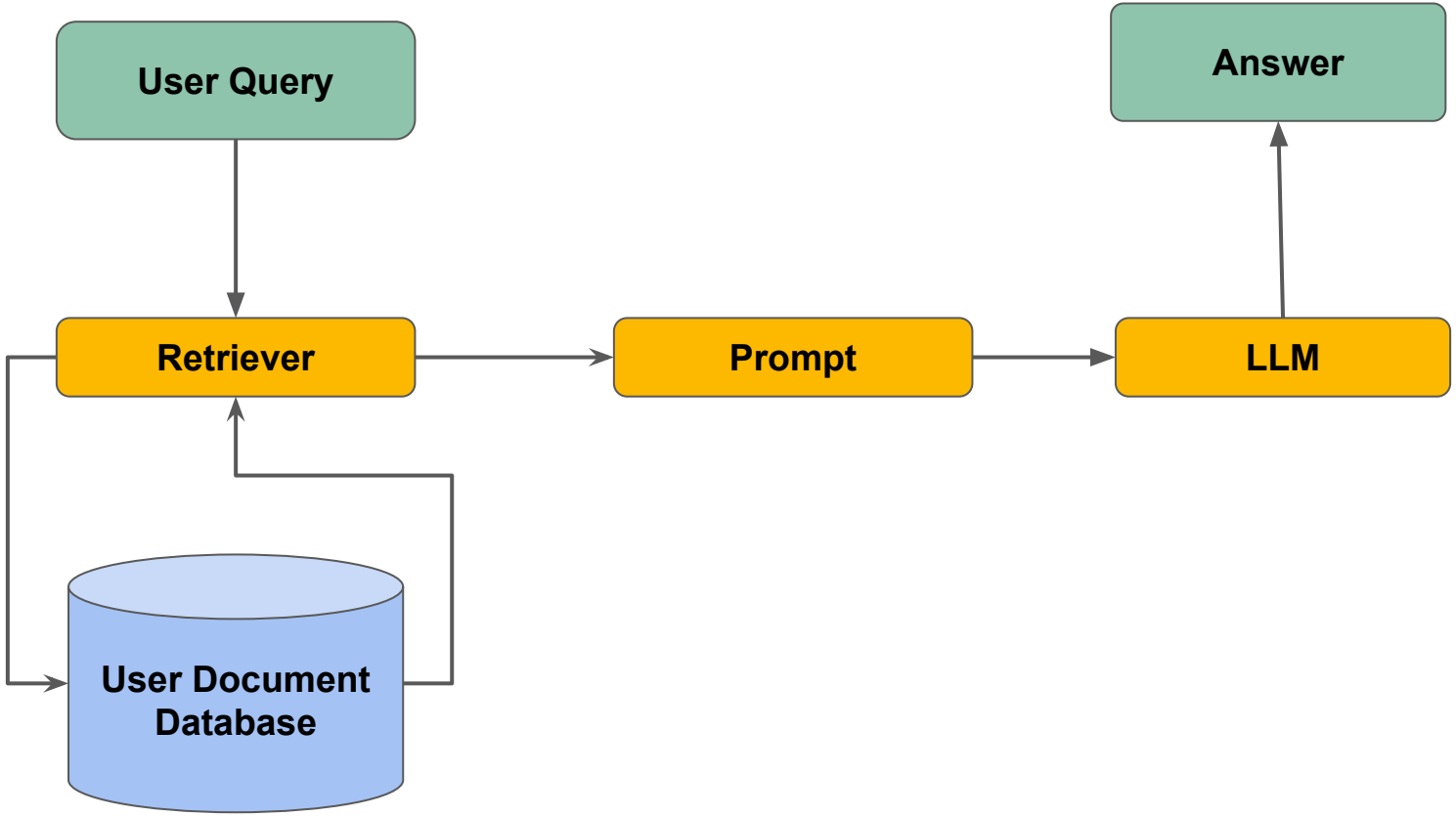


Without Docker:
Physical Memory: 16.00 GB
Memory Used 9.67 GB



With Docker:
Physical Memory: 16.00 GB
Memory Used: 15.34 GB

Interacting with RAG



Demo: Sample RAG Questions

Play Demo Video

```
What are my options for sick leave?

What happens on my first year anniversary?

When are time entries due?

What are regular office hours?
```

Competitor Analysis Project

Link to Claude.ai Analysis

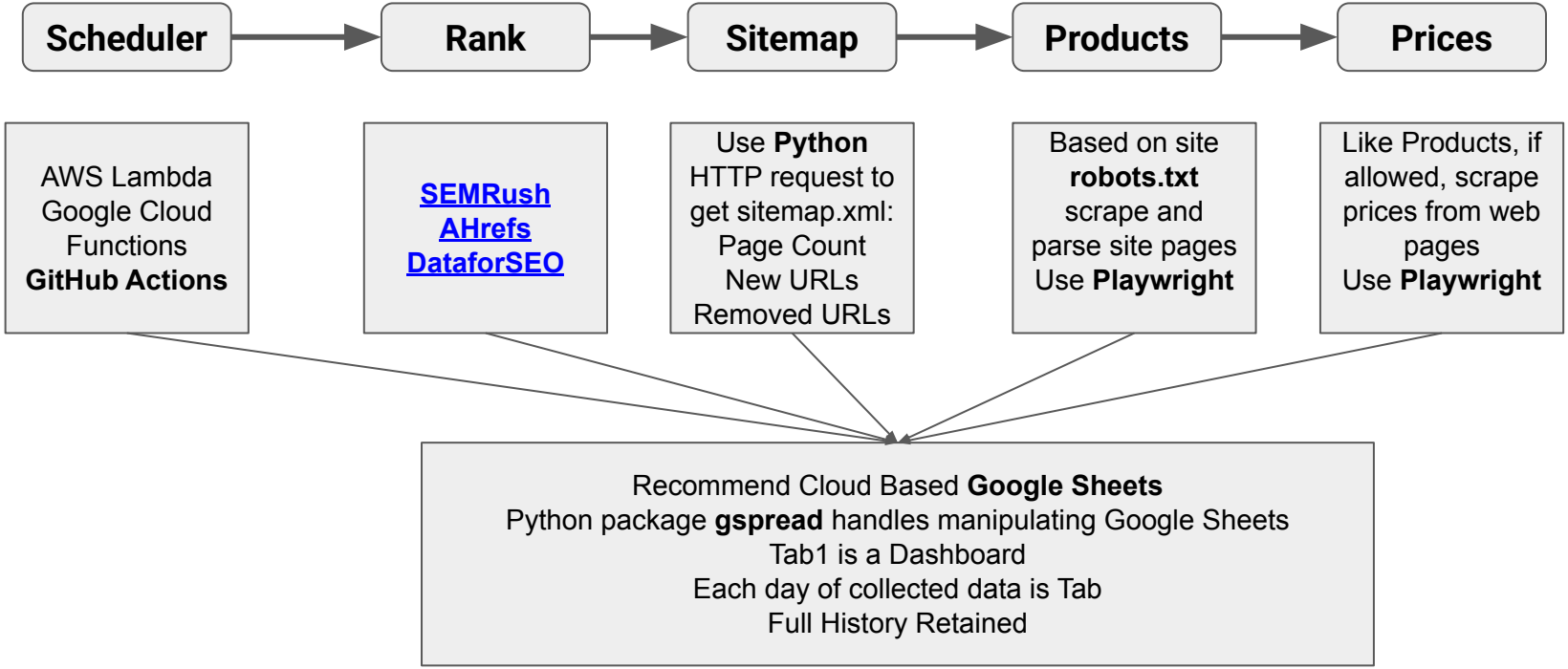
Proposal for Project

An AI **automation tool** that runs a **series of competitor analysis tasks** every day at midnight. Those tasks would include **rank tracking, sitemap analysis, product analysis, and price analysis**.

The results are logged in an **Excel** or **Google Sheets** document. Each day of results is logged in a new tab. Every day it runs and captures new data, the **AI runs an analysis that compares this day's data to previous days, weeks, months,** etc. and sends a **daily email report** summarizing it's findings to the marketing team.

Additionally, the first tab of the document is a **dashboard page** that displays a **summary of all collected data** with default preset graphs and comparisons. It also includes a **custom report builder** using elements like date range selection and category (sitemap, product, rank, etc.) so that you can make comparisons however you want, and the **AI would dynamically build present comparison bullet point analysis on the fly** as you change your custom report building.

Architecture: 1. Collect Data

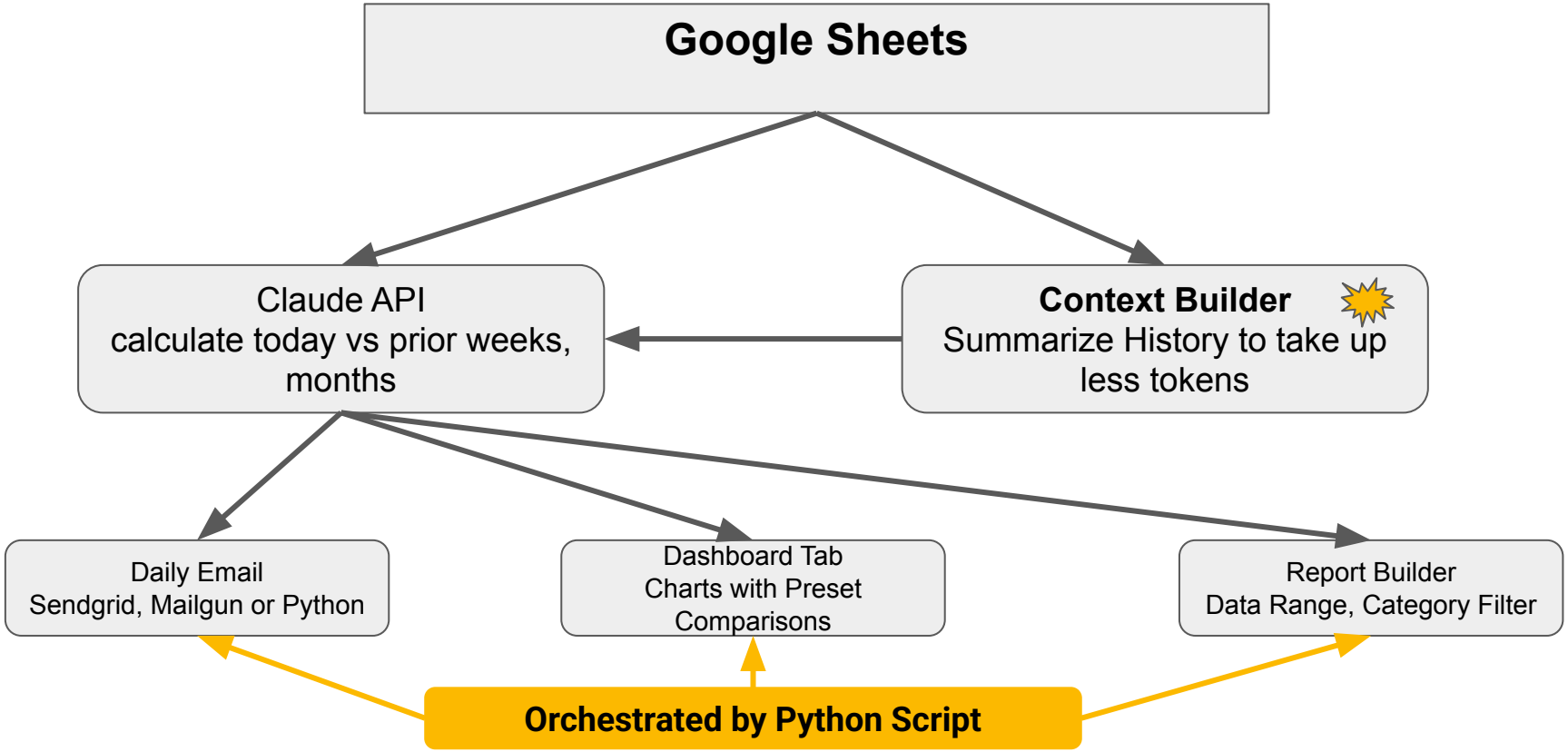


robots.txt

Block Crawlers	`User-agent: *` `Disallow: /`
Allow Crawlers Everything	`User-agent: *` `Allow: /`
Block Specific Files	`User-agent: Bingbot` `Disallow: /temp-files/`

Add Site Map	`Sitemap: https://www.example.com/sitemap.xml`
	`User-agent: *`
	`Disallow: /private/`

Architecture: 2. Storage



Architecture: 3. Analyze

Large historical datasets need summarization before being passed in, **otherwise token costs explode**.

Token Management

Scraping Ethics
robots.txt + Terms of Service

Dynamic AI User Interface

Prompt: `trend observations, anomalies, notable changes, and a plain-English summary`

Architecture: 4. Report

Python	script with many useful packages to help with data, email, and reports
gspread	Python package to interact with Google Sheet
Playwritght	Microsoft Testing tool that can help get information out of web page
Anthropic SDK	Post APIs to Anthropic for Claude processing
SendGrid	Cloud based tool for sending Emails
React/Next.js	JavaScript Framework for Creating Web Apps with UI and API

Projects

Project Ideas

- Create a Prompt Library that can be shared with other employees
- Create Customer Personas an a set of Social Media Posts for Each Person
- Create a set of Product and Seasonal Videos for Social Media
- Use AI Tools to find out trending topics and create company posts using the trends
- Create a Marketing plan spanning a month or a year
- Create content that Sales can share with customers or potential customers
- Use AI to create Dashboards using from marketing data
- Use NotebookLM to research a topic or help make a decision

Help with Projects

Contact me at rebeccapeltz@gmail.com for help with your projects

I'm available to help through April 17